

Matt Eisenhauer

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EDUCATION

Georgia Institute of Technology | MS, Computational Data Analytics (GPA: 3.85) | Expected May 2027

Syracuse University | BS, Information Management & Technology (GPA: 3.41) | May 2018

SKILLS

Data Science & ML: Python (Pandas, NumPy, Scikit-Learn, PyTorch), R, Predictive Modeling, Ensemble Methods (XGBoost, Random Forest), Regularization (LASSO, Ridge), Unsupervised Clustering (K-Means, GMM)

Statistical Analysis: Stochastic Modeling, Discrete-Event Simulation, Time-Series Forecasting, Experimental Design

Data Engineering & Visualization: SQL (Azure SQL / MS SQL Server), Data Pipelines (ETL / ELT), SQLAlchemy, Relational Database Design, Version Control (Git), Matplotlib, Seaborn, Plotly, ggplot2

PROJECTS

Stochastic Discrete-Event Simulation Model: In-Game Stolen Base Optimization

- Engineered a discrete-event simulation pipeline in Python that models stolen base attempts over 10,000 trials per matchup
- Ingested 2025 MLB Statcast data to create normal distributions for lead distance, pitcher delivery, and pop times
- Integrated an RE24 matrix framework to establish situational break-even success thresholds for "Go/No-Go" decisions
- Leveraged OpenCV for frame-by-frame video verification of raw game footage to validate pitcher time-to-plate distributions and backtest model accuracy

Shorten Up or Swing Away? A Machine-Learning Approach to Two-Strike Hitting

- Processed millions of rows of MLB Statcast data via pybaseball API, manipulating high-dimensional swing profile features
- Engineered a K-Means pipeline to cluster and identify distinct hitter archetypes based on two-strike swing profile delta
- Trained supervised regression models (XGBoost, Random Forest, Linear Regression) to predict xwOBA and evaluate traditional "shorten up" strategy vs. aggressive swingers

Predictive Modeling of Elite Biomechanical Performance (Driveline Open Biomechanics Project)

- Preprocessed 1,900+ Driveline athlete profiles, deploying a 5-NN imputation framework to resolve missing tracking data
- Engineered custom anatomical asymmetry metrics to evaluate physical range-of-motion variances and rotational efficiency
- Trained tree-based ensembles and regularized models (LASSO/Ridge) to isolate lower-body peak power metrics as primary drivers of pitch velocity and bat speed

WORK EXPERIENCE

UC Santa Barbara Baseball

Remote

Database Architect / Analytical Engineer (Freelance)

June 2024 - Present

- Engineered an automated ETL pipeline to ingest and process proprietary TrackMan data across NCAA Division 1
- Designed and deployed an Azure SQL database of normalized play-by-play data optimized for predictive modeling
- Provided support for data quality issues by implementing preprocessing validation scripts to proactively flag errors
- Optimized relational schemas and indexing, slashing query latency and analyst processing times by hours

Bank of America

New York, NY

Vice President – Infrastructure Engineer / Architect

January 2024 – Present

- Architected scalable infrastructure and networking solutions for high-volume global banking applications, ensuring system resilience and seamless data flow across enterprise platforms
- Developed Python scripts to optimize hostname / IP collection, removing manual lookups to map network architectures
- Designed application network architectures based on scale metrics, supporting file transfer baseline requirements

Consultant / AVP - Infrastructure Engineer

July 2018 – December 2023

- Created interactive data visualizations and dashboards to monitor application health and security events
- Spearheaded operational support for critical application outages, managing high-pressure real-time communications to senior stakeholders and authoring data-driven post-incident root-cause reviews